- Design modification
- Marker Making Systems(MMS)
- Pattern Design Software (PDS)
- Pattern Generation Software (PGS)
- Body Measurement Software (BMS)
- Texture Mapping
- Embroidery systems
- Specification and costing systems

15.3 Application of Cad in Textile And Garment Designing

The fashion business is growing rapidly and brings great opportunities and challenges to textile and clothing enterprises. Apparel industries follow the trend of the international market and latest technologies for industrial up gradation. Many textile and fashion designers use CAD systems for creating textile and garment designing. The designer may start by hand-drawing a few rough images. Then, the rough images are scanned into a computer and final designing is done using CAD.

A designer uses CAD to modify design drafts and make changes. Depending on the type of textile designing, a designer use different types of CAD software. There are systems for designing as well as creating knitted, printed fabrics, illustration and sketch pad applications that allow a person to draw freehand directly on the computer. There are CAD systems that show a designer how a fabric might drape for a particular style of garment. Some CAD programs even design embroidery patterns.

CAD finds its practical utility in textile, apparel and fashion industry right from design initiation and production stage through lay planning, spreading, pattern making, cutting and finally sewing. Two-dimensional pattern design software's based on flat patternmaking techniques have been successfully explored for patternmaking, grading, and creating pattern libraries within software for future retrieval.

However, a further progress made in the field is three-dimensional (3D) software. It works in virtual environment permitting 3D visualization of drape and fit on virtual model or avatar. This method eliminates exhaustive process of physical sample generation. 3D scanning, mapping, customized avatars in accordance with specific anthropometric and facial features, customized garments are some of the innovative and exciting avenues available with CAD software packages. Enhanced productivity, competitiveness, and shorter delivery schedules can be guaranteed by linking design, manufacturing operations along with other preproduction steps of patternmaking, grading, and marking.

DO YOU KNOW? Why Enterprise Resource Planning (ERP) Software in Textile and Apparel Industry?

It is software that collects information from all sections of the enterprise, assists in planning and implementation of various functions relating to all departments



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